

curriculum vitae of
Syed Ali Muqtadir

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EDUCATION

Oct 2022 – Sep 2026	PhD in Neuroscience Thesis title: The Heading Brain: Characterising the Acute Neurobiological Response to Football Heading Advisors: Dr Magdalena Ietswaart, Dr Colin Moran, Prof Lindsay Wilson Submitted: 30 September 2026	UNIVERSITY OF STIRLING, UNITED KINGDOM
Sep 2017 – Dec 2018	MSc in Neuroscience Thesis title: Effect of Transcranial Direct Current Stimulation on Probabilistic Learning in Patients with Schizophrenia Advisor: Dr Natasza Orlov	KING'S COLLEGE LONDON, UNITED KINGDOM
Sep 2012 – Dec 2016	BSc (Hons) in Psychology	UNIVERSITY OF CENTRAL PUNJAB, PAKISTAN

RESEARCH EXPERIENCE

Oct 2022 – Sep 2026	Doctoral Researcher - Completed a Medical Research Scotland-funded PhD project comprising three studies characterising the acute neurobiological response to controlled football heading. - Collected data from TMS-EEG, EEG/ERP and cognitive tests from 25 participants before and after heading and control exposure to index cortical E-I balance. - Acquired and analysed 7 T MRS data before and after heading at ICE, Glasgow and Forschungszentrum Jülich. - Investigated acute trajectory changes in plasma biomarkers of neuronal and glial injury (NfL, GFAP, UCH-L1, BD-tau) following heading exposure in a 24-hour serial blood-sampling protocol (n = 12, 10 time points) using ultra-sensitive immunoassays (Simoa) at the UK DRI, Edinburgh. - Initiated a collaboration with the School of Engineering, Edinburgh, supplying heading kinematic data for computational simulations of brain biomechanics (EdiFEHM). Collaborated with industry partner Magstim to test single-pulse and paired-pulse TMS protocols for the project. - Contributed to increasing technical capabilities at Stirling with the first use of TMS-EEG in the department and implementation of a low-cost navigated TMS system for coil and head tracking. - Secured additional funding through two successful grant applications. Managed ethics and governance approvals at three sites. Acquired certifications in phlebotomy, IV cannulation and MR safety.	UNIVERSITY OF STIRLING, UNITED KINGDOM
Aug 2019 – Aug 2022	Research Associate - Contributed to three research projects on online misinformation, social media communication and digital content virality. - Applied a logistic diffusion model to engagement data for 'viral' YouTube videos from a large public dataset to identify virality tipping points (Bhimani et al., 2024). - Collaborated on drafting a manuscript for a study on how the spread of anti-vaccine misinformation is modulated by certain cognitive heuristics (Ali et al., 2022).	LAHORE UNIVERSITY OF MANAGEMENT SCIENCES, PAKISTAN
Feb 2018 – Aug 2018	Graduate Researcher Investigated the effects of tDCS on cognitive function in patients with schizophrenia. Worked within the South London and Maudsley NHS Foundation Trust to administer tDCS protocols, PANSS interviews and cognitive assessments. This work resulted in two shared-first author publications (Scientific Reports, 2021; Psychiatry Research, 2022).	KING'S COLLEGE LONDON, UNITED KINGDOM

OUTPUTS

Authors who contributed equally are marked with a †.

PUBLICATIONS

1. Bhimani, A., Zainulabdin, K., Ali, K., **Ali Muqtadir, S.**, & Hausken, K. (2024). Social media virality: Reaching the tipping point. *Journal of International Business Research and Marketing*, 8(3), 27–41. <https://doi.org/10.18775/jibrm.1849-8558.2015.83.3003>
2. Ali, K., Li, C., Zain-ul-abdin, K., & **Muqtadir, S. A.** (2022). The effects of emotions, individual attitudes towards vaccination, and social endorsements on perceived fake news credibility and sharing motivations. *Computers in Human Behavior*, 134, 107307. <https://doi.org/10.1016/j.chb.2022.107307>
3. Orlov, N. D.[†], **Muqtadir, S. A.[†]**, Oroojeni, H., Averbeck, B., Rothwell, J., & Shergill, S. S. (2022). Stimulating learning: A functional MRI and behavioral investigation of the effects of transcranial direct current stimulation on stochastic learning in schizophrenia. *Psychiatry Research*, 317, 114908. <https://doi.org/10.1016/j.psychres.2022.114908>
4. Orlov, N. D.[†], Sanderson, J.[†], **Muqtadir, S. A.[†]**, Kalpakidou, A. K., Michalopoulou, P. G., Lu, J., & Shergill, S. S. (2021). The effect of training intensity on implicit learning rates in schizophrenia. *Scientific Reports*, 11(1), 6511. <https://doi.org/10.1038/s41598-021-85686-5>

PREPRINTS

5. **Muqtadir, S. A.**, Aston, T., Reid, K. R., Lember, L.-M., Di Virgilio, T. G., Keith, G., Hunt, D., Teixeira-Dias, F., Binkofski, F., Moran, C. N., & Ietswaart, M. (2026). Acute temporal dynamics of brain injury plasma biomarkers following controlled football heading. *bioRxiv*. <https://doi.org/10.64898/2026.09.09.749919>

IN PREPARATION

6. Patel, H. J.[†], **Muqtadir, S. A.[†]**, Choi, C.-H., Shaikh, U. J., Oros-Peusquens, A.-M., Shah, N. J., Moran, C. N., Ietswaart, M., & Binkofski, F. Neurometabolite changes following football heading: A 7 T magnetic resonance spectroscopy study.

PREREGISTRATIONS

- **Muqtadir, S. A.**, Aston, T., Moran, C., & Ietswaart, M. (2025). The effect of football heading on blood biomarkers of brain injury. *ISRCTN*. <https://doi.org/10.1186/ISRCTN44241334>
- **Muqtadir, S. A.** (2025). Impact of football heading on TMS-EEG indicators of neural excitation and inhibition. *OSF*. <https://doi.org/10.17605/OSF.IO/4AC6J>

FUNDING AND AWARDS

2022 – 2026

PhD Studentship

MEDICAL RESEARCH SCOTLAND

2025

Facility Access Funding Grant

RS MACDONALD CHARITABLE TRUST

2022

Early Career Exchange Fund

SINAPSE (SCOTTISH IMAGING NETWORK)

2013 – 2016

Academic Merit Scholarship

UNIVERSITY OF CENTRAL PUNJAB

SKILLS

Brain stimulation	TMS (single-pulse; paired-pulse SICI, LICI, ICF), concurrent TMS-EEG (TEP analysis), tDCS
Neuroimaging	7 T MRS (including edited MRS), structural MRI (volumetric analysis), EEG/ERP (acquisition, preprocessing, spectral analysis)
Physiology	EMG (motor-evoked potentials, cortical silent period), venepuncture and peripheral IV cannulation (certified), eye-tracking (Gazepoint), Near Point Convergence
Biological	Blood sample processing (plasma separation and storage), plasma biomarker quantification (Quanterix HD-X)
Assessment	Structured clinical interviews (PANSS), cognitive and neuropsychological assessments (CANTAB, Cogstate, WASI)
Programming & software	R (statistical analysis, data visualisation), MATLAB (EEGLAB, Osprey, SPM), FSL, FreeSurfer, BrainVision Analyzer, Biopac AcqKnowledge

TEACHING AND SUPERVISION

University of Stirling

Lecturing

- Delivered teaching for a final-year undergraduate module, “Advanced Methods in Human Physiology Research”, on using Transcranial Magnetic Stimulation (TMS) in research, including a practical demonstration.

Project and dissertation supervision

- Co-supervised a master’s project investigating EEG activity patterns and natural environment exposure, providing training in EEG data collection and analysis.
- Supervised a third-year group research project, guiding students through the research pipeline from design and ethical considerations to data analysis and writing.

Tutoring and small-group teaching

- Tutored final-year undergraduate students in workshops on writing study proposals based on real-world scenarios.

Laboratory-based teaching

- Led and supported practical laboratory sessions on TMS and EEG for psychology master’s students.
- Delivered a five-week training series on EEG/ERP analysis in BrainVision Analyzer for work-placement students and staff.

Assessment

- Marked a variety of assignments across several undergraduate modules, including Introductory Psychology, Developmental Psychology and Psychology in Context.

Lahore University of Management Sciences

- Trained undergraduate researchers in eye-tracking data collection using a Gazepoint system.
- Delivered teaching on research methods for undergraduate students.

PRESENTATIONS AND PUBLIC ENGAGEMENT

Mar 2026	Talk on ‘What does football heading do to the brain?’ for Scottish schools	MRS ACADEMY
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Mar 2024	Poster presentation at Researcher Development Event	MEDICAL RESEARCH SCOTLAND
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COURSES AND CERTIFICATIONS

Jul 2026	MRI Safety Training for MHRA Category B Staff	NHS SCOTLAND TURAS LEARN
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Mar 2025	Peripheral IV Cannulation	GEOPACE TRAINING
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Sep 2023	Visceral Mind: Functional Neuroanatomy Summer School	BANGOR UNIVERSITY
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Jun 2023	Phlebotomy	GEOPACE TRAINING
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